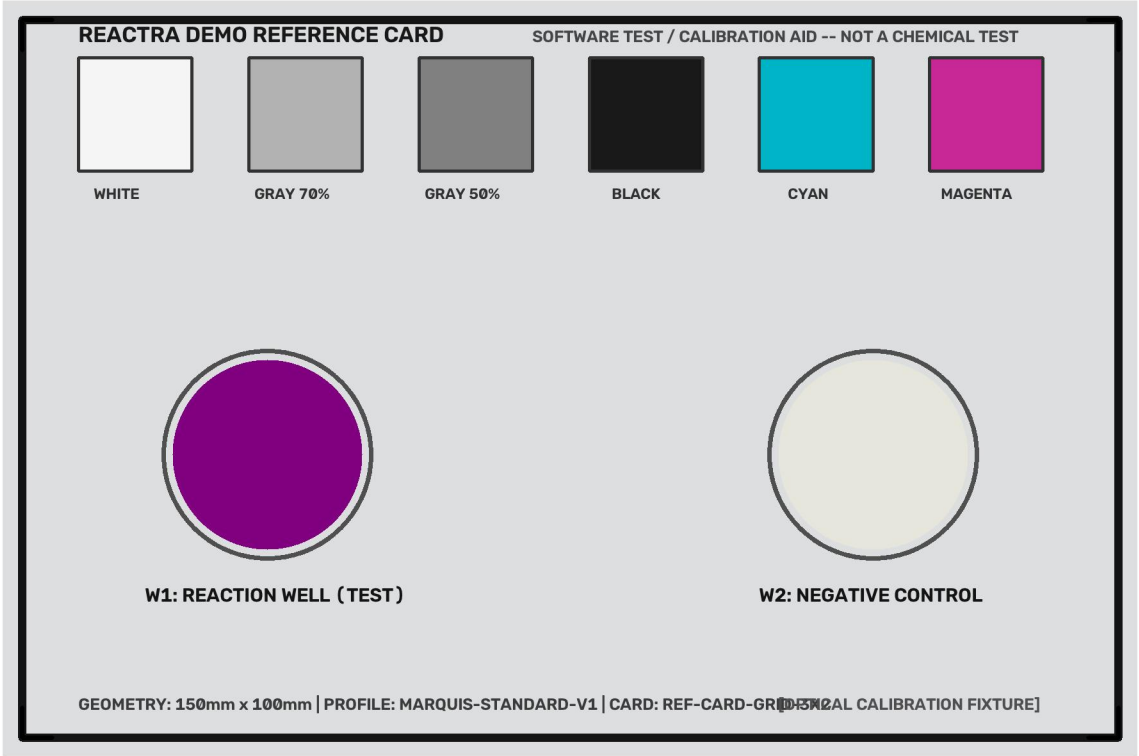




PHYSICAL SCALE VERIFICATION RULER (Must measure exactly 10.0 cm / 100 mm)



1. PRINTING & PHYSICAL PREPARATION

- Print Scale: Print at 100% scale / Actual Size. DO NOT select 'Fit to Page' or 'Shrink to Printable Area'.
- Paper Choice: Use heavy white matte paper or cardstock (200 - 300 gsm). Avoid glossy photo paper to prevent specular glare.
- Physical Size: Final trimmed card measures exactly 150 mm wide by 100 mm high (Aspect Ratio: 1.5).
- Cutting: Cut cleanly along the 4 corner crop marks. The solid black outer border must remain fully intact.
- Orientation: Place card flat on a neutral, non-reflective surface. Color patches and reaction wells must face camera.

2. CAMERA ALIGNMENT & ILLUMINATION PROTOCOL

- Camera Distance: Position camera 20 cm to 35 cm directly above the card so the card fills 40% - 60% of the viewport.
- Plane Alignment: Hold device plane parallel to card surface. Skew angle exceeding 20° is rejected by the Adaptive Capture Guard.
- Illumination: Use uniform, diffuse white illumination (indirect daylight or diffuse soft white LEDs).
- Anti-Glare Rule: Avoid direct camera flash, directional spotlights, or low-angle glare over reaction wells (Glare limit < 5%).
- Focus Sharpness: Ensure sharp focus before triggering capture. Laplace blur variance must exceed 1200 threshold.

MANDATORY NOTICE: OPTICAL TEST FIXTURE & CALIBRATION AID ONLY

This reference card is a software test fixture and colorimetric calibration aid designed for verifying camera acquisition and computer vision algorithms. It does NOT contain real chemical reagents and does NOT represent an actual chemical reaction.